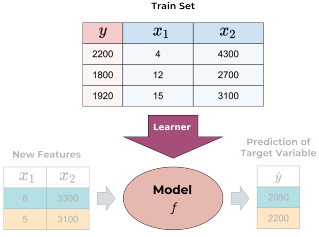
Introduction to Machine Learning

ML-Basics
Learner



Learning goals

- Know formal definition of learner
- Understand that a learner receives training data and outputs the best model from $\mathcal{H}$

SUPERVISED LEARNING EXAMPLE

Imagine we want to investigate how working conditions affect productivity of employees.

- It is a **regression** task since the target *productivity* is continuous.
- We collect data about worked minutes per week (*productivity*), how many people work in the same office as the employee in question, and the employee's salary.

Features x		Target y
People in Office (Feature 1) x_1	Salary (Feature 2) x_2	Worked Minutes Week (Target Variable)
4	4300 €	2220
12	2700 €	1800
5	3100 €	1920

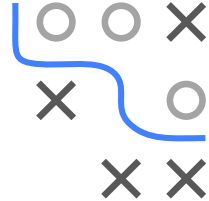
$n = 3$

$p = 2$

$x_1^{(2)}$ $x_2^{(1)}$ $y^{(3)}$

IDEA OF SUPERVISED LEARNING

Goal: Identify the functional relation that maps features to target.

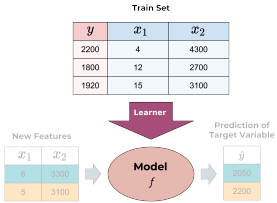


- **Supervised** learning means we use *labeled* data to learn model f.
- Later, we use model f to predict y for new *unlabeled* data.

	x_1	x_2	Functional Relationship	y
Already seen Data	4	4300 €	f	2200
	12	2700 €		1800
	15	3100 €		1920
New Data	6	3300 €		???
	5	3100 €		???

LEARNER DEFINITION

- Algorithm for finding f is called **learner** / **learning algorithm** / **inducer**.
- The learner is our means of picking the best element from the hypothesis space $\mathcal{H}$ for given training data.
- Formally, it maps training data $\mathcal{D} \in \mathbb{D}$ (plus a vector of **hyperparameter** control settings $\boldsymbol{\lambda} \in \boldsymbol{\Lambda}$) to a model:



$$\mathcal{I} : \mathbb{D} \times \boldsymbol{\Lambda} \to \mathcal{H}$$

- Practically, we often construct a mapping

$$\mathcal{I} : \mathbb{D} \times \boldsymbol{\Lambda} \to \Theta$$